the inclusion map we simply have $\mathfrak{b}^{\mathrm{c}} = \mathfrak{b} \cap A$. If $\mathfrak{a}$ is an A-ideal, in general $\varphi(\mathfrak{a})$ is not a B-ideal; but we can instead consider the B-ideal generated by $\varphi(\mathfrak{a})$, the *extension* of $\mathfrak{a}$ to B, sometimes denoted $\mathfrak{a}^{\mathrm{e}}$.

In the case of interest to us, A is an integral domain, $B = S^{-1}A$ is the localization of A with respect to some multiplicative set S, and $\varphi = \iota$ is injective, so we view A as a subring of B. We then have

$$\mathfrak{a}^{\mathrm{e}} = \mathfrak{a}B := (ab : a \in \mathfrak{a}, b \in B). \tag{2}$$

We clearly have $\mathfrak{a} \subseteq \varphi^{-1}(\varphi(\mathfrak{a})B) = \mathfrak{a}^{\mathrm{ec}}$ and $\mathfrak{b}^{\mathrm{ce}} = \varphi(\varphi^{-1}(\mathfrak{b}))B \subseteq \mathfrak{b}$; one might ask whether these inclusions are equalities. In general the first is not: if $B = S^{-1}A$ and $\mathfrak{a} \cap S \neq \emptyset$ then $\mathfrak{a}^{\mathrm{e}} = \mathfrak{a}B = B$ and $\mathfrak{a}^{\mathrm{ec}} = B \cap A$ are both unit ideals, but we may still have $\mathfrak{a} \subsetneq A$. However when $B = S^{-1}A$ the second inclusion is an equality; see [1, Prop. 11.19] or [2, Prop. 3.11] for a short proof. We also note the following theorem.

Theorem 2.1. *Let S be a multiplicative subset of an integral domain A. There is a one-to-one correspondence between the prime ideals of $S^{-1}A$ and the prime ideals of A that do not intersect S given by the inverse maps $\mathfrak{q} \mapsto \mathfrak{q} \cap A$ and $\mathfrak{p} \mapsto \mathfrak{p}S^{-1}A$.*

Proof. See [1, Cor. 11.20] or [2, Prop. 3.11.iv]. □

Remark 2.2. An immediate consequence of (2) is that if $a_1, \ldots, a_n \in A$ generate $\mathfrak{a}$ as an A-ideal, then they also generate $\mathfrak{a}^{\mathrm{e}} = \mathfrak{a}B$ as a B-ideal. As noted above, when $B = S^{-1}A$ we have $\mathfrak{b} = \mathfrak{b}^{\mathrm{ce}}$, so every B-ideal is of the form $\mathfrak{a}^{\mathrm{e}}$ (take $\mathfrak{a} = \mathfrak{b}^{\mathrm{c}}$). It follows that if A is noetherian then so are all its localizations, and if A is a PID then so are all of its localizations.

An important special case of localization occurs when $\mathfrak{p}$ is a prime ideal in an integral domain A, and $S = A - \mathfrak{p}$ (the complement of the set $\mathfrak{p}$ in the set A). In this case it is customary to denote $S^{-1}A$ by

$$A_{\mathfrak{p}} := \{a/b : a \in A, b \notin \mathfrak{p}\}/\sim, \tag{3}$$

and call it the *localization of A at $\mathfrak{p}$*. The prime ideals of $A_{\mathfrak{p}}$ are then in bijection with the prime ideals of A that are contained in $\mathfrak{p}$. It follows that $\mathfrak{p}A_{\mathfrak{p}}$ is the unique maximal ideal of $A_{\mathfrak{p}}$ and $A_{\mathfrak{p}}$ is therefore a local ring (whence the term *localization*).

Warning 2.3. The notation in (3) makes it tempting to assume that if a/b is an element of $\operatorname{Frac} A$, then $a/b \in A_{\mathfrak{p}}$ if and only if $b \notin \mathfrak{p}$. This is not necessarily true! As an element of $\operatorname{Frac} A$, the notation "a/b" represents an equivalence class; thus even if $b \in \mathfrak{p}$, if $a/b = a'/b'$ with $b' \notin \mathfrak{p}$, then $a/b \in A_{\mathfrak{p}}$. As a trivial example, take $A = \mathbb{Z}$, $\mathfrak{p} = (3)$, $a/b = 9/3$ and $a'/b' = 3/1$. You may object that we should write a/b in lowest terms, but when A is not a unique factorization domain it is not clear what this means.

Example 2.4. For a field k, let $A = k[x]$ and $\mathfrak{p} = (x-2)$. Then

$$A_{\mathfrak{p}} = \{f \in k(x) : f \text{ is defined at } 2\}.$$

The ring A is a PID, so $A_{\mathfrak{p}}$ is a PID with a unique nonzero maximal ideal (the ideal $\mathfrak{p}A_{\mathfrak{p}}$), hence a DVR. Its maximal ideal is

$$\mathfrak{p}A_{\mathfrak{p}} = \{f \in k(x) : f(2) = 0\}.$$

The valuation on the field $k(x) = \operatorname{Frac} A$ corresponding to the valuation ring $A_{\mathfrak{p}}$ measures the order of vanishing of functions $f \in k(x)$ at 2. The residue field is $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \simeq k$, and the quotient map $A_{\mathfrak{p}} \twoheadrightarrow A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ sends f to $f(2)$.